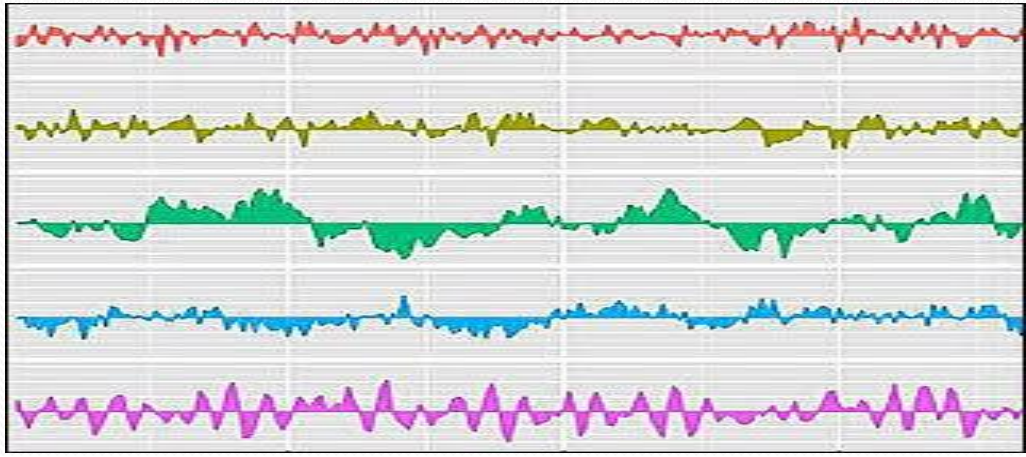
TIME SERIES ANALYSIS – STD 714



UTG
Knowledge, Truth & Development
Established 1999

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Part A: Class Work

- Introduction to Time Series Analysis
- Stationary time series
- autocovariance functions
- Strict and weak stationarity
- White noise and linear processes
- Autoregressive (AR)
- Moving Average (MA)
- Autoregressive Moving Average (ARMA) processes

Part A: Class Work Contd.

- Parameter estimation methods
- Diagnostic testing and goodness-of-fit
- Prediction and forecasting
- Non-stationary time series and integrated processes
- Deterministic and stochastic trends

Part A: Class Work Contd.

- Autoregressive Integrated Moving Average (ARIMA) models
- Unit root tests:
  - Dickey–Fuller
  - Augmented Dickey–Fuller
  - Phillips–Perron tests

  Stationarity testing: KPSS test
- Estimation and prediction of non-stationary time series
- Seasonal ARIMA (SARIMA) models

Part B:Laboratory Work

- Plotting and visualizing time series data
- Estimating AR, MA, ARMA, and ARIMA models
- Performing unit root and stationarity tests
- Forecasting future values using fitted models
- Comparing forecasting accuracy measures
- Seasonal decomposition of time series data
- Mini-projects using real economic or financial datasets

Required Software for Lab. Work





Installing Base-R: http://cran.r-project.org/bin/windows/base

Installing Rstudio: http://www.rstudio.com/products/rstudio/download/

Introduction to Time Series Analysis (TSA)

What is time series?

- Data obtained from observations collected sequentially over time
- Extremely common across all disciplines
- Key Feature: Observations are NOT independent - unlike classical statistics
- We study models that incorporate dependence

Examples across Disciplines

- Business: weekly interest rates, daily stock prices, monthly price indices, yearly sales
- Meteorology: daily high/low temperatures, annual precipitation, hourly wind speeds
- Agriculture: annual crop & livestock production, annual soil erosion, annual export sales
- Biological Sciences: electrical activity of heart [millisecond intervals]

PURPOSE OF TSA

- To Understand / Model: The stochastic mechanism that gives rise to the observed series
- To Predict / Forecast: Future values based on history of the series and other related factors

COMPONENTS OF A TIME SERIES

- A time series Y_t is assumed to be made up of 4 components
- Idea: $Y_t = \mathrm{f}(T_t, S_t, C_t, I_t)$
- Decomposition can be:
  - Additive or
  - Multiplicative

Components of a TS Contd.

Trend Component

- Definition: Long-term movement / direction of the series
- Upward, downward or stationary over long period
- Caused by: Population growth, technological progress, inflation
- Example: Increasing population, long-term rise in sales
- Duration: Many years

Components of a TS Contd.

Seasonal Component

- Definition: Regular, periodic fluctuations that repeat within a year
- Fixed period: Daily, weekly, monthly, quarterly
- Caused by: Climate, customs, festivals, calendar
- Example: Umbrella sales in rainy season, high air travel in December
- Duration: Less than 1 year, regular pattern

Components of a TS Contd.

Cyclical Component

- Definition: Long-term up and down movements around the trend
- NOT fixed period, unlike seasonal
- Caused by: Business cycle - boom, recession, depression, recovery
- Example: Economic cycles, business expansion and contraction
- Duration: 2 to 10 years, irregular

Components of a TS Contd.

Irregular / Random Component

- Definition: Unpredictable, residual fluctuations
- What remains after removing Trend, Seasonal, Cyclical
- Caused by: Random events - wars, floods, strikes, pandemic
- Example: Sudden drop due to COVID-19
- Also called: Noise, Error component

Graphical ilustration TS Components



Time Series Model

Additive Model

- It assumes components are independent
- $Y_t = T_t + S_t + C_t + I_t$
- Used when seasonal variation is constant

Multiplicative Model

- Most economic series
- $Y_t = T_t \; X \; S_t \; X \; C_t \; X \; I_t$
- Used when seasonal variation increases with trend

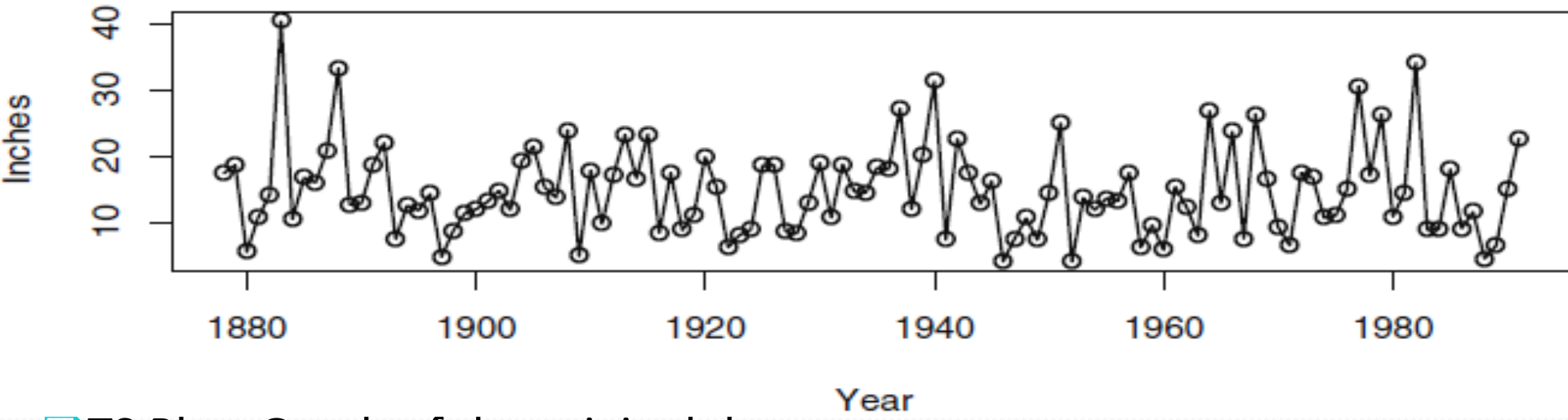
UNIQUE FEATURES OF TS MODELS

- Classical assumption: Observations arise independently from a common population - FAILS in time series
- Current value depends on past values
- Models incorporate dependence

Case Study: Annual Rainfall in Los Angeles

- Data: Annual rainfall amounts in Los Angeles, California
- Period: Over 100 years
- From 1880 - 1980

TS plot for Annual Rainfall in Los Angeles



- TS Plot: Graph of the original data
- X-axis: Year [1878 - 1992]
- Y-axis: Rainfall in Inches [0 - 40]
- Each point = Total rainfall for one year
- Connected by line to show evolution over time

KEY FEATURES OF THE PLOT

- High variability - No stable mean visually
- Range: From <10 inches [dry years] to 40 inches [1883 wettest]
- No obvious long-term trend or seasonality
- Outlier year: 1883 stands out clearly
- Series appears stationary but volatile

Purpose of Time Series Plot

- First step in any time series analysis: Visualize
- To check for:
  - Trend
  - Outliers
  - Variation
  - Possible dependence
- Foundation before plotting ACF and Lag-plot

Interpretation For Modeling

- Not white noise - variation is large
- Suggests need for descriptive statistics: Mean ~ 15 inches, SD ~ 6 inches
- Next question: Is this year's rainfall useful to predict next year's? -> Requires Lag Plot

Interpretation For Modeling Contd.

White Noise

- White noise e_t is PURE randomness.
- No pattern left.
- Mean = 0
- Variance = Constant
- No autocorrelation:
- Corr(e_t , e_{t-k})=0 for all k≠0
- ACF plot = all spikes inside bounds
- Ljung-Box test p-value > 0.05
- White noise = Nothing more to explain.
- Model is complete.

Interpretation For Modeling Contd.

Not White Noise

- Residual / series still has structure.
- It is NOT random.
- It means:There is still autocorrelation - past values predict future values
- There is still information left in the data you have not modelled
- The model is incomplete / inadequate
- On ACF/PACF plot: You will see significant spikes outside the confidence bounds
- Ljung-Box test p-value < 0.05 - reject white noise

The 3 Main Required Steps

- Model Specification (Identification)
- Model Fitting (Estimation)
- Model Diagnostics (Checking)

We cycle through these steps until an acceptable model is found

Model Specification / Identification

- Goal: Select tentative class of models appropriate for observed series
- How:
  - Look at time plot
  - Compute many statistics: ACF, PACF, EDA
- Important: Model chosen is TENTATIVE - subject to revision

Tentative Models To Consider

If series is STATIONARY

- AR(p) Model: PACF cuts off after p, ACF tails off
  - Use when present depends on past values
- MA(q) Model: ACF cuts off after q, PACF tails off
  - Use when present depends on past errors
- ARMA(p,q) Model: Both ACF and PACF tail off
  - Mix of AR and MA

Tentative Models To Consider Contd.

If series is NON-STATIONARY

- ARIMA(p,d,q) Model: Need differencing
  - d = 1 if trend, ACF decays very slowly
  - Plot shows trend upward/downward
- SARIMA(p,d,q)(P,D,Q)s: Seasonal
  - ACF shows spikes at lag s, 2s, 3s (e.g. lag 12 for monthly)
  - Time plot shows repeating seasonal pattern

Tentative Models To Consider Contd.

If series is STATIONARY BUT VOLATILE

- ARCH / GARCH Model: Mean is stable but variance clusters
  - ACF of squared residuals significant

- ARMA + GARCH: For both mean and volatility

Tentative Models To Consider Contd.

If series is WHITE NOISE

- No Model needed: $Y_t = \mu + e_t$
- No pattern, pure random

PRINCIPLE OF PARSIMONY

- Definition: Use smallest number of parameters that will adequately represent the series
- Quote - Albert Einstein:

"Everything should be made as simple as possible but not simpler" - Parzen (1982)

- Avoid over-parameterization. Fitting a model with too many parameters e.g using AR(5) when AR(1) is enough

Thank you